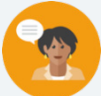


2.2 Square Values and Square Roots

Lesson Introduction

 Benchmarks	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real-number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line. MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.		 Learning Targets	- I can determine the positive and negative solutions for $x^2 = p$, where p is a whole number. - I can use the vocabulary related to radicals. - I can interpret and evaluate a square root.
 Benchmark Coherence	Building On N/A		Working Toward MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system. MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.	
 Essential Questions	- How many solutions are there for equations in the form $x^2 = p$, where p is a whole number? - What is the square root of p, where p is a whole number? - How are square roots and square numbers related?			
 Lesson Narrative	In this lesson, students apply their conceptual understanding of perfect squares from Lesson 1 to discover how to find the square roots of perfect squares by using inverse operations. Students expand their vocabulary as they learn to identify the square root symbol, radical, radicand, and index.			
 Component Summary	2.2.1 Warm-Up (~5 min.)	Estimate and justify estimation (SE p. 69)	2.2.4 Guided Instruction (10 min.)	Introduce radicals and square roots (SE p. 71)
	2.2.2 Guided Instruction (10 min.)	Learn to solve equations in the form $x^2 = p$ (SE p. 69)	2.2.5 Guided Practice (5-10 min.)	Practice solving equations involving squares and evaluating square roots (SE p. 72)
	2.2.3 Try It (5 min.)	Find solutions to equations in the form $x^2 = p$ (SE p. 70)	2.2.6 Your Turn! (5 min.)	Additional practice solving equations involving squares and evaluating square roots (SE p. 73)
	2.2.7 Wrap-Up (~5 min.)	Complete two statements to reflect on the lesson (SE p. 73)		
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Square root - Perfect square - Radical - Radicand - Index	(Optional) Chart paper & markers		N/A

 Teacher Tips	Common Misconceptions - Students may not put negative numbers in parentheses when squaring, resulting in an incorrect value. - Students may initially think the inverse operation of squaring is dividing by two, based on a misconception of squaring involving multiplication by two.	Things to Consider - It is important to emphasize that each square root has both a positive and negative value throughout Lesson 2. Students learn about the principal square root in Lesson 4 as they approximate square roots of nonperfect squares. - Students will apply the knowledge and skills they develop in this lesson in Unit 6 using the Pythagorean Theorem to determine missing side lengths of right triangles. - If time is an issue, then consider assigning 2.2.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 2.2.4 Guided Instruction presents students with new vocabulary and notation. To foster students’ metacognitive and metalinguistic awareness as they work with this new vocabulary and notation, prompt students to use the language in their discussions and label the equations using the new terms.	MA.K12.MTR.3.1: Mathematical Fluency Students develop fluency by practicing the skills they acquire, first with supports in 2.2.2 Guided Instruction and 2.2.5 Guided Practice, and then independently in 2.2.3 Try It and 2.2.6 Your Turn!. This practice will help them flexibly and accurately find and interpret square roots of whole numbers.
 Differentiate Instruction	If students struggle with the new vocabulary and notation, consider using different colors to highlight the vocabulary and the corresponding notation that it represents. Provide frames to help students remember to use parentheses, such as: $p = (_)$ and $_ = x^2$. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

2.2.1 Warm-Up: Estimation

Component Overview: Students use the growth rate of deep-sea clams to estimate a clam's size.



2.2.1 Warm-Up: Estimation

1. In 2006, scientists in North Wales found a deep-sea clam that was 507 years old. Deep-sea clams have the slowest growth rate, growing as little as 8.4 millimeters (mm), or one-third inch, every 100 years.



- a. Estimate the clam's size when the scientists found it in 2006.

Answers will vary. The clam was about 42 mm. The clam was about $1\frac{2}{3}$ inches.

- b. Explain your reasoning for your estimation.

Answers will vary. I multiplied 8.4×5 because it grows 8.4 mm per 100 years and 100 goes into 500 five times. I divided 500 by the 100 years it takes to grow $\frac{1}{3}$ inch and got 5. I then multiplied $\frac{1}{3} \times 5$.



MA.K12.MTR.2.1: Multiple Representations
Encourage students to use creative methods to represent their reasoning using models or diagrams. For example, a student might illustrate their understanding of thirds using a tape diagram or bar model, showcasing a proficiency in fractions compared to only multiplying 8.4 millimeters by the number of years.

2.2.2 Guided Instruction: Solving Equations in the Form $x^2 = p$

Component Overview: Students learn to solve equations in the form $x^2 = p$, where p is a whole number. Students conclude that when $p \neq 0$, there are two solutions, one positive and one negative. Students also discover that there are no solutions when p is negative.



2.2.2 Guided Instruction: Solving Equations in the Form $x^2 = p$

1. Consider the equation $x^2 = 1$.
 - a. Discuss with your partner what value(s) make the equation true.
1 and -1 will both make the equation true.
 - b. Find the solutions for $x^2 = 1$
 $1 \cdot 1 = 1$ and $-1 \cdot -1 = 1$
 $1^2 = 1$ and $(-1)^2 = 1$
 - c. Show the solutions that satisfy the equation.
 $1 \cdot 1 = 1$ and $-1 \cdot -1 = 1$
 $1^2 = 1$ and $(-1)^2 = 1$
2. Find the solutions for $64 = m^2$.
8 and -8
3. Find the solution for $x^2 = 0$.
0



Students are not expected to solve by finding the square root. Rather, they should apply their knowledge of perfect squares to determine the answers. Ask students to consider what number(s), when squared, are equal to _____. Press students to think about how opposites may help them solve for x .

Most students will think of the positive solution, but they may not initially think of the negative solution unless prompted.



Provide frames to help students remember to use parentheses, such as:

$$p = (_) \quad \text{and} \quad _ = x^2.$$

2.2.2 Guided Instruction: Solving Equations in the Form $x^2 = p$ (continued)

For equations in the form $x^2 = p$, where p is a whole number, there are always two solutions, except in the case where $p = 0$. For all other whole number values of p , there is one positive and one negative solution.

4. Consider the equation $x^2 = -36$.

- a. Discuss with your partner whether you think it is possible to find a solution to this equation.
Answers will vary. It is not possible because if you multiply two positive numbers the answer is positive and when you multiply two negative numbers the answer is also positive. It is not possible because the result of squaring a number is always positive.

b. Complete the statements.

There ☐ is ☒ is not a real number that can be

multiplied by itself to result in -36 . Therefore,

- ☒ there is not a real solution to the equation $x^2 = -36$.
- ☐ the real solution(s) to the equation $x^2 = -36$ is(are) _____.



Ask students to consider with their partner why there are not two solutions to the equation $x^2 = 0$ but there are for all other equations where p is positive.

0 is neither positive nor negative. While other whole numbers can result from multiplying a positive value by itself or a negative value by itself, the only value that can result in 0 when multiplied by itself is 0.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
 Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.



Consider asking questions to probe student thinking during the discussion such as:

- What is the sign of the product of two positive numbers? Two negative numbers?
- Is there any number that can be multiplied by itself to get a negative number?

2.2.3 Try It: Solving Equations in the Form $x^2 = p$

Component Overview: Students find the positive and negative solutions of $x^2 = p$, where p is a positive whole number.



2.2.3 Try It: Solving Equations in the Form $x^2 = p$

Find the solution(s) for each equation.

1. $x^2 = 36$
 $6, -6$

2. $t^2 = 144$
 $12, -12$

3. $49 = n^2$
 $7, -7$



MA.K12.MTR.3.1: Fluency

Fluency does not mean speed. Rather, fluency means flexibility in problem-solving and making decisions for what efficiency means for each student. Prompt questions on students' process rather than their product. How did you know it was 6? Why is it 6 and -6 ?

2.2.4 Guided Instruction: Square Roots

Component Overview: Students learn that the inverse of squaring a number is finding the square root. Students learn the vocabulary and notation that is used when working with radicals and explore the values of square roots.



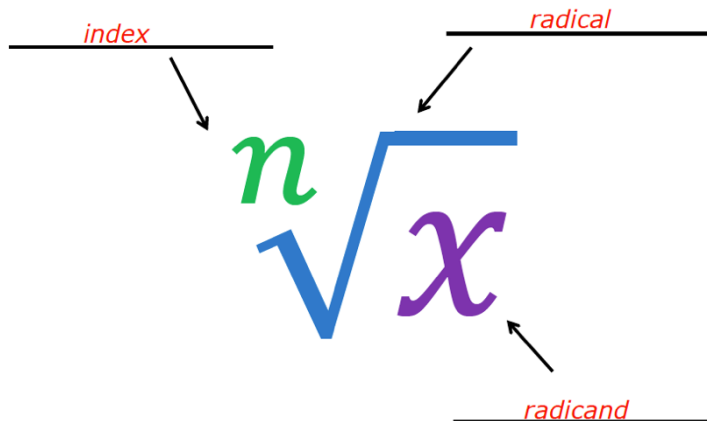
2.2.4 Guided Instruction: Square Roots

A perfect square is the product that results from squaring a number. Squaring a number is multiplying it by itself. The inverse operation to squaring a number is taking its **square root**.

The symbol $\sqrt{\quad}$ is called a **radical**. It is used to represent a **square root**. The value under the radical sign is called the **radicand**.

The **index** is the number outside the radical sign that indicates the root being taken.

1. Use this information to label the parts of a radical expression.



The symbol $\sqrt{\quad}$ is interpreted as the square root, unless an index other than 2 is indicated.



If students struggle with the new vocabulary and notation, consider use different colors to highlight the vocabulary and the corresponding notation that it represents. Consider making an anchor chart for reference.



Clarify that when finding the square root, the index is 2, but no index is written. Connect this to not writing an exponent when expressing numbers to the first power and not writing a coefficient of 1 in front of a variable. In the next lessons, students will be working with an index of three, and these are called cube roots.

Explain that the index can be any rational number.

2.2.4 Guided Instruction: Square Roots (continued)

2. Consider the equations shown.

$$(4)^2 = 16$$

$$(-4)^2 = 16$$

The equations show which values can be squared to result in 16. Based on the information given about square roots, what is $\sqrt{16}$?

4 and -4

As explored in the previous lesson with the graph of $y = x^2$, every number has both a positive and negative square root, except for 0. The $\pm$ symbol is read, "plus or minus," and is used in front of the radical to denote both roots.

3. Complete each statement.

a. $\sqrt{4} = \pm 2$ because $\underline{2} \times \underline{2} = \underline{4}$ and

$$\underline{-2} \times \underline{-2} = \underline{4}$$

b. $\sqrt{121} = \pm \underline{11}$ because $\underline{11} \times \underline{11} = \underline{121}$

$$\text{and } \underline{-11} \times \underline{-11} = \underline{121}$$

4. Explain why there is not a real number equivalent to $\sqrt{-81}$.

Answers will vary. It is not possible because if you multiply two positive numbers, the answer is positive, and when you multiply two negative numbers, the answer is also positive. It is not possible because the result of squaring a number is always positive.



Students learn in 6th grade that when squaring a negative, it should be put in parentheses to show that the negative is also being squared. This may be a misconception students have, though. Question 2 provides an opportunity to address this.

Consider showing students the following to clarify:

- $-4^2 = -16$ because it is the same as $-(4 \times 4)$.
- $(-4)^2 = 16$ because it is the same as (-4×-4) .



Compare and Connect

To foster students' metacognitive and metalinguistic awareness as they work with this new vocabulary and notation, prompt students to use the language in their discussions and label the equations using the new terms.



- When working with square roots, what is the index?
- How many square roots does ____ have?
- What symbol is used to represent both roots?



Explain that the square roots of negative numbers are nonreal numbers called imaginary numbers. Tell students that they will learn about imaginary numbers in future courses, but for now, they are only looking for real solutions or indicating that there is no real solution.

2.2.5 Guided Practice: Squares and Square Roots

Component Overview: Students recognize that finding square roots and squaring a number are inverse operations. Students solve equations by applying the inverse operations of squaring a number or/and evaluating square roots.



2.2.5 Guided Practice: Squares and Square Roots

- Xavier completed two problems as shown.

Find the solution(s) for $b^2 = 25$.

$b = -5$ and $b = 5$

What is the value of $\sqrt{25}$? 5

- Discuss Xavier's answers with your partner.
- Complete the statements.

Xavier answered ☐ one problem ☐ both problems correctly.

There ☐ is only one solution ☐ are two solutions to the equation

$b^2 = 25$ because there ☐ is only one value ☐ are two values that

make(s) the equation true. When finding the value

of a square root, consider what value can be

multiplied by itself to result in the radicand.

Therefore, the value of $\sqrt{25}$ is ☐ 5. ☐ -5. ☐ ± 5 .



Monitor as students complete question 1 to listen to their discussions and check that they correctly complete the statements with their partner.

Consider pausing the class after they have finished question 1 to debrief it together, if needed, based on the observed discussions.



When selecting which option represents the square root of 25, students would be correct if they selected the first two choices or the third. The goal is for them to recognize the third choice as the notation used to represent both the positive and negative values of the square root and the positive and negative solutions to the equation $x^2 = p$. Prompt students to think about why the third choice would be the best option, given that both 5 and -5 are correct options.

2.2.5 Guided Practice: Squares and Square Roots (continued)

2. Find the value of each of the following.

a. $\sqrt{81} = \pm 9$

b. $\sqrt{225} = \pm 15$

c. $\sqrt{9} = \pm 3$

d. $\sqrt{-64} = \text{no real solution}$

3. Solve each equation.

a. $a^2 = 169$
 $\sqrt{a^2} = \sqrt{169}$
 $a = \pm 13$

b. $225 = x^2$
 $\sqrt{225} = \sqrt{x^2}$
 $\pm 15 = x$



Students will be introduced to the principal square root in a future lesson as they begin working to approximate square roots of nonperfect squares. It's essential, though, that they understand that each square root has both a positive and negative value, so that should be reinforced throughout the lesson.



Introduce the notation for applying the inverse operation to solve the equation. Students will use this notation in Unit 6 when solving for missing side lengths in right triangles using the Pythagorean Theorem as well as in future courses.

- “To isolate a , take the square root of a^2 . Take the square root of 169 as well to maintain equivalence in the equation. The square root of a^2 is equal to a because the two operations (squaring and taking the square root) undo each other. They are inverse operations.”



Reinforce this thinking as students continue practicing by asking the following questions, as needed.

- What are inverse operations? What is the relationship between squaring a number and finding the square root?
- How can you isolate the variable?
- If you apply the square root to the side of the equation with the variable raised to the second power, what do you need to do to the other side of the equation?

2.2.6 Your Turn!

Component Overview: Students independently practice finding the square roots of positive whole numbers and solving equations by applying inverse operations.



2.2.6 Your Turn!

1. Find the value of each of the following.

a. $\sqrt{49} = \pm 7$

b. $\sqrt{196} = \pm 14$

2. Solve each equation.

a. $w^2 = 4$
 $\sqrt{w} = \sqrt{4}$
 $w = \pm 2$

b. $121 = k^2$
 $\sqrt{121} = \sqrt{k}$
 $\pm 11 = k$



What is your strategy for finding square roots? Do you memorize or have an algorithm with unique connections?

Monitor for students using unique or creative methods for finding square roots. Elicit these methods whole group to highlight and encourage flexibility in problem-solving.

2.2.7 Wrap-Up: Reflection

Component Overview: Students respond to two out of five prompts of their choice as they reflect on their learning. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



2.2.7 Wrap-Up: Reflection

1. Choose **two statements to complete** as you reflect on today's lesson.
 - a. One thing that made the most sense to me today was ...
Answers will vary. That the square root of a number is both a positive and a negative number. That the inverse of squaring a number is to find the square root.
 - b. One thing that I just don't understand is ...
Answers will vary. I don't understand how to find the square root of a very large number. I don't understand how there is no real solution to the square root of a negative number.
 - c. One thing I learned today was ...
Answers will vary. I learned that finding the square root of a number means to find both the positive and negative values that results in the radicand when squared. I learned the sign $\pm$ means positive and negative.
 - d. One thing I still wonder about is ...
Answers will vary. How do you find the square root of numbers that are not perfect squares? What is the index outside of the radical sign used for?
 - e. One thing I noticed about square roots is ...
Answers will vary. The answer is both a positive and negative number. It is the inverse of squaring a number.



Consider allowing students to use an audio recorder (such as Flip Grid) or speech-to-text software.